

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 01 · LECTURE NOTES

When the Input Has Interests

What changes when the data is reported rather than given.

MODULE OUTCOMES

- Explain why strategic inputs change correctness.
- Give concrete examples where the shift matters.
- Distinguish the analysis and design questions.
- State the two ways out of the problem.
- State this course's position on mechanism claims.

1 The shift

EVERY ALGORITHMS COURSE BEFORE THIS ONE ASSUMED THE INPUT WAS GIVEN; HERE IT IS REPORTED BY SOMEBODY WITH A PREFERENCE OVER YOUR OUTPUT

A sorting algorithm's input does not care how it is sorted — a bidder's valuation does care who wins the item, and will be reported accordingly if misreporting helps, which is not dishonesty so much as rational participation.

So an algorithm that is provably correct on the reported input may be useless with respect to the true one — and the true input is not observable, by anybody, ever, **which is the whole difficulty of the subject** and what distinguishes it from robustness or error tolerance.

Which means correctness has to be redefined: a mechanism is correct if it produces the right outcome *given that agents will report strategically* — which is a strictly stronger requirement than being correct on honest input, and is not implied by it.

And the cleanest route to it is to make truthful reporting optimal for each agent individually — incentive compatibility — so that the reported input simply is the true one and the ordinary algorithmic analysis applies again (Module 06, and it is why that module is the course's centre).

2 Where this bites

System	The reported input	The incentive to misreport
Advertising auction	Bids.	Pay less than your value; win the slots that are profitable.

System	The reported input	The incentive to misreport
Cloud or cluster scheduler	Job priority and declared resource requirements.	Overstate both, always — see the note.
Peer review assignment	Bids on papers, and review scores.	Advance one's own work and collaborators; a real and studied problem.
School choice and residency matching	Ranked preference lists.	Rank strategically rather than truly, to get a better assignment (Module 09).
Inter-domain routing	Advertised path costs and availability.	Attract paying transit traffic, or avoid carrying it free.
Search and recommendation signals	Links, reviews, engagement.	Manufacture the signal — CSCE 670 Module 11's adversarial information retrieval.

Any shared resource allocated by reported need has this structure — which includes most scheduling, most quota systems, and every priority field a user can set. The cloud scheduler row is the one most engineers have already met without naming it: the priority field that is always 'high', the resource request that is always double what is used.

AND THE FAILURE MODE IS FAMILIAR EVEN WHERE THE THEORY IS NOT

Every priority field that users set for themselves ends up uniformly high — which is a mechanism design failure with a well understood cause, and is usually patched by imposing quotas rather than by redesigning the rule, which works and is a different thing from solving it.

And every ranking signal that can be manufactured eventually is — which is CSCE 670 Module 11's adversarial information retrieval, and is exactly the same problem with a different vocabulary: a metric that determines an allocation becomes a target (CSCE 676 Module 13's proxy problem).

So the subject is not exotic: it is the theory behind failures that engineers meet constantly and usually treat as abuse rather than as a design defect — which matters, because the two framings lead to completely different responses.

Which is the framing this course takes: if the rules reward misreporting, then the misreporting is the system working as specified — and the fix is the rules, not enforcement, not education, and not an appeal to good behaviour.

3 Two questions

- **The analysis question: given these rules, what will happen? — which needs a solution concept to predict behaviour (Module 02) and a way to measure the damage relative to what a designer would have chosen (Module 04's price of anarchy).**
- **And the design question: given a desired outcome, what rules produce it? — which is mechanism design (Module 06) and is the engineering half of the subject, and the half this course is ultimately for.**
- **Analysis occupies Modules 02 through 05, and it is largely bad news: equilibria are hard to compute (Module 03) and are worse than the social optimum (Module 04), sometimes substantially.**
- **Design occupies Modules 06 through 11, and it is largely good news: a great deal can be implemented truthfully, including some things that look impossible at first, with named exceptions.**
- **And the exceptions are theorems (Module 06 §4's Myerson-Satterthwaite, Module 10 §3's Gibbard-Satterthwaite) — which tell you what to stop trying to build, which is the most practically valuable thing a negative theorem does and is worth the effort of understanding precisely.**

4 The position

A MECHANISM'S GUARANTEES HOLD UNDER A MODEL OF THE AGENTS, AND THE AGENTS ARE PEOPLE

'Truthful reporting is a dominant strategy' assumes that agents know their own valuations, can compute best responses, and do not collude — all three of which fail sometimes, and the first is less obvious than it sounds: people frequently do not know what something is worth to them.

And the second-price auction is the standard example: provably truthful under the model, and bidders in practice still shade their bids, because they do not believe the rule, do not understand it, or are optimising against a repeated interaction the single-shot model omits (Module 07 §4).

Which does not make the theory useless — it makes it a model whose assumptions should be stated alongside its conclusions, exactly like every other model in this program (CSCE 640 Module 13's three things, CSCE 628 Module 13's five).

So the rule here: name the solution concept, name the agent model, and name the impossibility you chose to live with — three things, and Module 13 is about all three.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Roughgarden — Twenty Lectures, lecture 1 (free)	§§1 and 3 — the framing, with the sponsored search example developed as the motivating case.
Nisan et al., chapter 1 (free PDF)	§3's two questions, and the book's organisation follows that division.
Roughgarden — Algorithmic Game Theory, CACM (free)	§§1 and 2 — a short argument for why computer scientists need this, aimed at exactly this course's audience.
Varian — Position auctions (free preprint)	§2's first row — the ad auction analysed, which is the deployment that made this field commercially serious.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the assumption this course removes**, and what it changes about correctness.
- 2 **Find three systems you use** whose input is reported by interested parties.
- 3 **For each, name the incentive to misreport.**
- 4 **Find a priority field** in a system you have access to, and look at its distribution.
- 5 **Describe one case** where misreporting is treated as abuse rather than as a design defect.
- 6 **Classify five questions** as analysis or design.
- 7 **State what incentive compatibility would buy you**, in one sentence.
- 8 **Find a quota that patches an incentive problem**, and say what the designed fix would be.
- 9 **List the three assumptions** behind a dominant-strategy claim.
- 10 **Find a case where one of them visibly fails.**

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What assumption does this course remove?
- 2 Why does that change the definition of correctness?
- 3 What is incentive compatibility for?
- 4 Give four systems with strategic inputs and their incentives.
- 5 Why is the priority-field failure a design defect?
- 6 State the analysis question and what it needs.

- 7 State the design question and which modules cover it.
- 8 Which half is bad news, and why?
- 9 What do the impossibility results give you?
- 10 Name the three assumptions behind a truthfulness claim.